Hierarchical Shell Manifold Theory after Foundational Closure:

Structural Maturity, Selection Incompleteness, and a Six-State Analog Benchmark

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Abstract

Hierarchical Shell Manifold Theory (HSMT) is a graded geometric and operator framework in which effective physics is obtained by projection from a larger structured state space. This paper records the theory's status after an extended sequence of structural audits, selector searches, and controlled benchmark design. The principal conclusion is deliberately two-sided. First, the mathematical architecture is mature under its present assumptions: the graded arena, associative operator laboratory, projection machinery, effective reduction, and downstream dimensional diagnostics can be stated coherently without identifying the grade label with dimension. Second, the same frozen assumptions do not uniquely determine the physical carrier rank, physical sector, state/boundary weighting, or a unique physical generator. Several elegant rank-three relations exist, including the balance $\dim \Lambda^1(V^*) = \dim \Lambda^2(V^*)$, but these become selectors only after an independent physical premise is added. We therefore formulate a scoped selection-incompleteness result for the audited selector class rather than a universal impossibility theorem.

A six-state analog benchmark, denoted P3, is derived as the next empirical step. For the path-graph Laplacian K_3 with positive eigenvalues 1 and 3, two candidate realizations yield the distinct modal laws $\omega_S(\lambda) = g\lambda$ and $\omega_I(\lambda) = g\sqrt{\lambda}$. The corresponding parameter-free frequency-ratio predictions are $R_S = 3$ and $R_I = \sqrt{3}$. A predeclared RevC laboratory protocol is specified, including independent mode verification, controlled preparation with retrospective validation, direct analog readout of the two modal channels, and a strict no-retuning rule. No experimental result is claimed. The role of P3 is to test whether the declared operator laws can be physically instantiated and experimentally discriminated in real hardware while leaving open the deeper question of which law, if any, nature selects.

1 Introduction

The purpose of this paper is not to announce physical closure of HSMT. Its purpose is to state precisely what has been achieved, what remains underdetermined, and what the next empirically meaningful step is.

The HSMT program began from the idea that physical descriptions may be organized by a graded family of manifold sectors rather than by a single manifold with one fixed dimensional interpretation. Subsequent development refined the framework into a cleaner operator architecture: a graded arena, an associative state/operator laboratory, controlled intergrade couplings, projection onto an effective sector, and downstream diagnostics of scale-dependent effective dimension. This refinement forced an important distinction: the hierarchy label N is not itself a spatial dimension, and an HSMT shell is not assumed to be an ordinary spatial shell.

A second maturation occurred when the framework was subjected to an extended selector search. Numerous mechanisms can make rank three mathematically distinguished, but none of the audited mechanisms uniquely selects the physical rank, sector, measure, and generator from the frozen core without an additional premise. This is a structural result about the present theory, not a defect to be hidden.

The paper therefore has three goals:

(i) state the mature HSMT architecture in compact form;

(ii) formulate the present selection-incompleteness result with explicit scope and firewall conditions; and

(iii) derive and predeclare the P3 six-state analog benchmark that provides the next laboratory-level test of the operator dynamics.

The September HSMT framework remains the architectural spine of the program. The present paper is a follow-up status and benchmark paper, not a replacement for that framework.

2 Frozen HSMT Architecture

2.1 Graded arena

The underlying HSMT arena is represented schematically as

$$\mathcal{M} = \bigsqcup_N \mathcal{M}_N, \tag{1}$$

where N is a hierarchy or grade label. The theory does not identify N with an observed spatial dimension. A useful distinction is

$$N \neq d, \tag{2}$$

where d denotes a dimensional characterization. More generally, HSMT keeps separate several notions that need not coincide:

$$d_{\mathrm{geom}}, \qquad d_{\mathrm{res}}, \qquad d_{\mathrm{spec}}, \qquad d_{\mathrm{dyn}}. \tag{3}$$

No equality among these quantities is assumed by definition.

2.2 Operator chain and effective reduction

The mature conceptual chain is

$$\mathcal{M} \longrightarrow (D_\rho, D_{\mathrm{sp}}) \longrightarrow \text{states/packets} \longrightarrow \langle N|M\rangle \longrightarrow \mathcal{D} \longrightarrow P_N \longrightarrow H_{\mathrm{eff}}. \tag{4}$$

At the minimal abstract level, the frozen core can be summarized as

$$\boxed{(\mathcal{H}, \{P_N\}, \mathcal{D})}. \tag{5}$$

Here $\mathcal{H}$ is the associative Hilbert-space laboratory, $\{P_N\}$ is the family of grade projectors, and $\mathcal{D}$ is a controlled reference operator. The reference operator is not automatically identified with a unique physical master operator.

The framework admits standard operator-control methods, including bounded-coupling self-adjointness arguments of Kato–Rellich type and exact projected resolvent/Feshbach constructions where their hypotheses are satisfied [2, 3]. These tools establish mathematical control of the laboratory and of projected dynamics; they do not by themselves identify the physically realized sector.

2.3 Dimensional characterization is downstream

The measurement layer is deliberately placed after the operator architecture. A generic scale-dependent diagnostic is

$$Q_N(\ell) \longrightarrow d_N(\ell) = \frac{d \ln Q_N}{d \ln \ell}. \tag{6}$$

The role of Q_N must be declared in each application. Thus a shell is not defined by inserting a dimensional value. Instead, dimensional behavior is inferred from the scaling of an observable or operator-derived quantity.

Definition 1 (HSMT shell)**.** *An HSMT shell is a dimensional regime of the underlying graded manifold, characterized by the scaling of accessible geometric, dynamical, or spectral degrees of freedom. Its effective dimension may be integer or fractional and may run with scale.*

3 Structural Maturity under the Present Assumptions

The present framework has passed a sequence of internal consistency and architecture checks. The resulting status can be summarized as follows.

1. The graded arena and effective projection structure are well-defined at the framework level.
2. Distinct notions of dimension are separated rather than conflated.
3. Associative state/operator constructions support controlled calculations.
4. Self-adjointness and projected-resolvent methods are available under explicit hypotheses.
5. Norm-preserving closed evolution may be imposed as an optional dynamical axiom. It is not derived from the frozen core, and its adoption does not automatically identify the evolution parameter with physical time, quantum mechanics, or Lorentzian spacetime.
6. Candidate many-body, support, overlap, spectral, and projected dynamics can be constructed without forcing a unique physical realization.

These results justify calling the current HSMT architecture *structurally mature under its present assumptions*. They do not justify calling the theory physically complete.

4 The Physical-Selection Problem

4.1 An instructive rank-three relation

Let V be a local carrier space of dimension q. Then

$$\dim \Lambda^1(V^*) = q, \qquad \dim \Lambda^2(V^*) = \frac{q(q-1)}{2}. \tag{7}$$

If one postulates a balance

$$\dim \Lambda^1(V^*) = \dim \Lambda^2(V^*), \tag{8}$$

then

$$q = \frac{q(q-1)}{2}, \tag{9}$$

so

$$q(q-3)=0. \tag{10}$$

For positive nonzero carrier rank,

$$\boxed{q=3}. \tag{11}$$

The same conclusion follows if one assumes the existence of an invertible local derivative-free exchange

$$J_0 : \Lambda^2(V^*) \to V^*. \tag{12}$$

Invertibility requires equality of dimensions and therefore again selects $q = 3$.

These are mathematically clean observations. They are not yet physical derivations, because HSMT does not presently derive the balance condition or the exchange law from the frozen core.

4.2 Audited selector class

The selector search considered a broad set of candidate mechanisms, including:

- rank/cardinality balance;
- $\Lambda^1 \leftrightarrow \Lambda^2$ matching;
- Hodge-type and Clifford-type exchanges;
- chirality and index-style arguments;
- topological or anomaly-inspired constraints;
- relation-curvature exchange mechanisms;
- compactification resistance and local-to-global completion;
- symmetry protection and larger symmetry embeddings;
- renormalization-group stability;
- ground-state, harmonic, or $\mathcal{D}^2$ selection;
- commuting-selector and positive-selector constructions;
- boundary/state weighting and history measures;
- bootstrap/self-consistency criteria.

Across this audited class, the recurring obstruction is the same: a mechanism either fails to select rank three uniquely, or it selects rank three only after a new structural/dynamical premise is inserted, or it leaves the physical measure/generator underdetermined.

4.3 Scoped selection-incompleteness result

Let F denote the presently frozen HSMT core assumptions, and let $\mathfrak{S}$ denote the audited selector class summarized above.

Proposition 1 (FSNG-0: scoped selection incompleteness)**.** *Within the presently frozen HSMT core F and the audited selector class $\mathfrak{S}$, no selector has been established that uniquely derives, without an additional independent physical premise,*

$$q_{\text{phys}} = 3, \tag{13}$$

while simultaneously fixing the physical sector, physical state/boundary weighting, and a unique physical generator.

Remark 1. *This statement is intentionally scoped. It is not a theorem that no possible future physical principle can select rank three. It states that the frozen theory and the audited selector mechanisms have not done so.*

The practical conclusion is

$$\boxed{\text{HSMT is structurally mature but physically selection-incomplete.}} \tag{14}$$

5 The Missing Physical Law

HSMT presently contains mathematically consistent candidate dynamics. What it does not yet contain is an independently justified fundamental law that determines which realization is physically selected.

The missing information may be represented schematically as a map

$$\mathfrak{P} : \text{admissible HSMT realizations} \longrightarrow \text{physically realized sector}, \tag{15}$$

where $\mathfrak{P}$ would need independent justification and empirical consequences.

At present, the theory does not uniquely derive all of the following:

$$\boxed{\text{physical sector}, \quad q_{\text{phys}}, \quad \rho_{\text{phys}}, \quad H_{\text{phys}}} \tag{16}$$

where ρ_{phys} denotes the physically relevant state/boundary weighting and H_{phys} the physical generator.

This separation is methodologically useful. It prevents an attractive mathematical identity from being silently promoted into a law of nature.

5.1 Selector firewall

Any future proposed selector should satisfy the following conditions before it is promoted to physical law:

1. The new premise must be declared explicitly.
2. The no-go or incompleteness result it evades must be identified.
3. The premise must have noncircular justification independent of the desired answer.

4. Rank three must be selected, not merely permitted.
5. The physical weighting/measure must also be fixed or separately justified.
6. Compatibility with the adopted dynamical and compositional structure must be checked.
7. The proposal must produce an independent falsification channel.

6 P3: A Six-State Analog Benchmark

The next high-value empirical step is not another internal selector search. It is a controlled physical implementation test of two declared operator laws.

6.1 Graph Laplacian

Let

$$K_3 = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}. \tag{17}$$

Its spectrum is

$$\mathrm{spec}(K_3) = \{0, 1, 3\}. \tag{18}$$

A convenient eigenbasis is represented by

$$\lambda_0 = 0: \qquad (1, 1, 1), \tag{19}$$
$$\lambda_1 = 1: \qquad (1, 0, -1), \tag{20}$$
$$\lambda_3 = 3: \qquad (1, -2, 1). \tag{21}$$

The six real state variables are collected as

$$Y = (u_1, u_2, u_3, v_1, v_2, v_3). \tag{22}$$

6.2 S realization

The S realization is

$$\dot{u} = gK_3 v, \qquad \dot{v} = -gK_3 u. \tag{23}$$

Therefore

$$\ddot{u} = -g^2 K_3^2 u. \tag{24}$$

For a mode of eigenvalue λ,

$$\omega_S(\lambda) = g\lambda. \tag{25}$$

For the two positive modes,

$$\boxed{R_S = \frac{\omega_S(3)}{\omega_S(1)} = 3.} \tag{26}$$

6.3 I realization

The I realization is

$$\dot{u} = gv, \qquad \dot{v} = -gK_3u. \tag{27}$$

Hence

$$\ddot{u} = -g^2K_3u, \tag{28}$$

and therefore

$$\omega_I(\lambda) = g\sqrt{\lambda}. \tag{29}$$

The positive-mode ratio is

$$\boxed{R_I = \frac{\omega_I(3)}{\omega_I(1)} = \sqrt{3} \approx 1.732050808.} \tag{30}$$

6.4 Modal readouts

The primary analog readouts are

$$X_1 = u_1 - u_3 \tag{31}$$

and

$$X_2 = \frac{u_1 - 2u_2 + u_3}{3}. \tag{32}$$

These isolate the two positive graph modes up to normalization. The experimental observable is

$$\boxed{R = \frac{f(X_2)}{f(X_1)}.} \tag{33}$$

The direct analog frequency ratio is the primary P3 observable. A comparator/latch/counter layer producing digital event outputs may be used as a secondary hardware diagnostic, but it is not required to define the primary benchmark.

6.5 Preparation state

The intended preparation is proportional to

$$u(0) \propto (1, -1, 0), \qquad v(0) \approx 0. \tag{34}$$

For

$$u = (A, -A, 0), \tag{35}$$

one finds

$$X_1 = A, \qquad X_2 = A. \tag{36}$$

Thus equal positive readout amplitudes provide a convenient modal preparation check.

This check is not a unique certification of the complete state. Adding any constant zero-mode component $c(1, 1, 1)$ leaves both X_1 and X_2 unchanged. This is acceptable for the primary benchmark because the zero mode does not alter the two nonzero modal frequencies. The preparation criterion is therefore modal rather than a unique full-state certification.

7 RevC Implementation and Predeclared Experimental Protocol

7.1 Laboratory controls

The RevC implementation uses ±12 V analog rails, a +5 V logic rail, and common ground. The MODE control is independently verified before each trial:

$$\text{MODE} \approx +5\ \text{V} \Rightarrow S, \qquad \text{MODE} \approx 0\ \text{V} \Rightarrow I. \tag{37}$$

In PREP, the preparation-routing network connects

$$\text{THN} \rightarrow \text{DIN1}, \qquad \text{THP} \rightarrow \text{DIN2}, \qquad \text{GND} \rightarrow \text{DIN3}, \tag{38}$$

with nominal settings

$$\text{THN} = -0.500\ \text{V}, \qquad \text{THP} = +0.500\ \text{V}. \tag{39}$$

Because the U-state integrators are inverting, this preparation drives the relative state toward $(+, -, 0)$.

7.2 Critical HOLD/PREP ordering

The reset/HOLD paths discharge the integrator feedback capacitors. Therefore a nonzero U-state cannot be established while U HOLD remains asserted. The correct order is:

$$\text{PREP} \rightarrow \text{release U HOLD} \rightarrow \text{short preparation interval} \rightarrow \text{RUN} \rightarrow \text{release V HOLD immediately.} \tag{40}$$

The V bank remains held near zero during preparation.

With nominal input resistance near $97.6\,\text{k}\Omega$ + trim and feedback capacitance 100 nF, a 0.500 V preparation input gives an order-of-magnitude ramp rate

$$\left|\frac{du}{dt}\right| \sim \frac{0.500}{RC} \sim 50\ \text{V/s}. \tag{41}$$

Accordingly, the preparation interval is of order milliseconds rather than seconds.

7.3 Retrospective preparation validation

To avoid a human reaction-time bias, the operator does not attempt to switch at an exact live amplitude. Instead, the complete oscilloscope acquisition is saved and the preparation state is assessed afterward.

For the first formal run, the predeclared engineering screening window is

$$0.3\ \text{V} \leq X_1, X_2 \leq 1.0\ \text{V} \tag{42}$$

and

$$\frac{|X_1 - X_2|}{\text{mean}(X_1, X_2)} < 0.05 \tag{43}$$

when evaluated at the PREP→RUN transition.

These bounds are predeclared engineering acceptance criteria, not HSMT predictions and not quantitatively derived from a measured noise or offset budget of the specific board. They are declared before inspection of the frequency-ratio result and therefore cannot be tuned after the fact.

7.4 Formal S-mode sequence

The formal S-mode procedure is:

1. Apply +12 V, −12 V, +5 V, and common ground.
2. Assert U HOLD and V HOLD.
3. Select S mode.
4. Verify the MODE node is approximately +5 V relative to ground.
5. Verify THN = −0.500 V and THP = +0.500 V.
6. Set PREP/RUN to PREP.
7. Connect/arm the oscilloscope on X_1 and X_2.
8. Keep V HOLD asserted.
9. Release U HOLD.
10. Allow a short preparation interval, nominally 5–15 ms.
11. Switch PREP→RUN.
12. Release V HOLD immediately as the next operator action.
13. Apply no additional external drive.
14. Record and save the autonomous X_1 and X_2 waveforms.
15. After the trial, inspect the saved traces at the PREP→RUN transition.
16. Accept or reject the preparation using only the predeclared engineering window.
17. For an accepted trial, record $f(X_1)$ and $f(X_2)$ and compute $R_S = f(X_2)/f(X_1)$.
18. Report raw frequencies, ratio, preparation amplitudes, instrument settings, and propagated measurement uncertainty.

The I-mode run is identical except that MODE is independently verified near 0 V and the ideal prediction is $R_I = \sqrt{3}$.

7.5 No-retuning firewall

The benchmark is predeclared. After observing a result, the operator must not alter MODE, preparation voltages, integrator gains, modal readout definitions, or acceptance rules in order to move the ratio toward the prediction. Any failed preparation is rejected by the predeclared preparation criterion, not by the measured value of R.

No arbitrary fixed pass/fail band is imposed on the final frequency ratio in advance of an uncertainty analysis. The exact ideal predictions are stated before the run, and the measured ratios are to be reported together with propagated experimental uncertainty.

8 What P3 Can and Cannot Establish

A successful controlled run satisfying

$$R_S \approx 3, \qquad R_I \approx \sqrt{3} \tag{44}$$

within experimental uncertainty would establish that a real analog apparatus can physically instantiate and experimentally discriminate the two declared operator laws.

It would *not* establish that:

- HSMT is a fundamental theory of nature;
- nature selects S rather than I;
- rank three has been fundamentally derived;
- the physical state/boundary measure has been selected;
- the HSMT shell ontology has been empirically confirmed.

This distinction motivates the evidence ladder in Table 1. A successful P3 run is an E1 result. It is

Table 1: Evidence ladder used to prevent overclaim.

Level	Interpretation
E0	Mathematical consistency and derivation inside the declared assumptions.
E1	Engineered analog or laboratory implementation of a declared HSMT operator structure.
E2	Unbiased analog emergence not built directly into the apparatus.
E3	Independent physical observation connected to a distinctive HSMT prediction.
E4	Fundamental evidence favoring HSMT over competing physical frameworks.

valuable precisely because its evidential scope is explicit. It is not an E2 result, because the operator laws are deliberately engineered into the apparatus.

9 Discussion

The present HSMT status is unusual but scientifically clean. The framework has become sufficiently mature that further internal elaboration is no longer automatically progress. In particular, repeated attempts to derive a physical selector from the same frozen assumptions risk converting a desired answer into a hidden premise.

The correct next step is therefore bifurcated.

On the mathematical side, work may continue on proofs, conditional consequences, spectral structure, dimensional diagnostics, and more demanding benchmarks. Such work should be labeled correctly as development within an assumption set.

On the physical side, genuinely new progress requires new input: a new independently motivated law, a microscopic dynamical principle, an external mathematical constraint with physical

justification, or empirical information. P3 supplies empirical information of a limited but important kind: whether the candidate operator laws can be realized and distinguished in actual hardware under a predeclared protocol.

The deeper selector problem remains open regardless of the P3 result. A successful P3 run would justify proceeding to stronger experiments. A failed run would also be scientifically useful, because it would identify either an implementation error, an unmodeled analog effect, or a deficiency in the declared hardware realization. The no-retuning firewall is therefore essential.

10 Conclusion

The current HSMT program has reached a clear checkpoint.

First, the structural architecture is mature under its present assumptions. The graded arena, associative operator laboratory, projection structure, effective reduction, and downstream dimensional characterization form a coherent framework.

Second, physical closure has not been achieved. The frozen theory does not yet contain an independently justified law that uniquely selects the physical rank, sector, weighting, and generator. Rank-three identities exist, but they become physical selectors only after additional premises are introduced.

Third, this incompleteness does not make further work directionless. It sharpens the next task. The P3 six-state benchmark produces two exact parameter-free modal ratio predictions,

$$\boxed{R_S = 3, \qquad R_I = \sqrt{3},} \tag{45}$$

which can be tested in a controlled analog implementation without claiming more than the experiment can establish.

The present paper therefore records a transition in the HSMT program: from architecture-building and internal selector search to controlled empirical discrimination. The theory is not physically complete, but its present incompleteness is now localized, explicit, and experimentally actionable.

A Compact Architecture Diagram

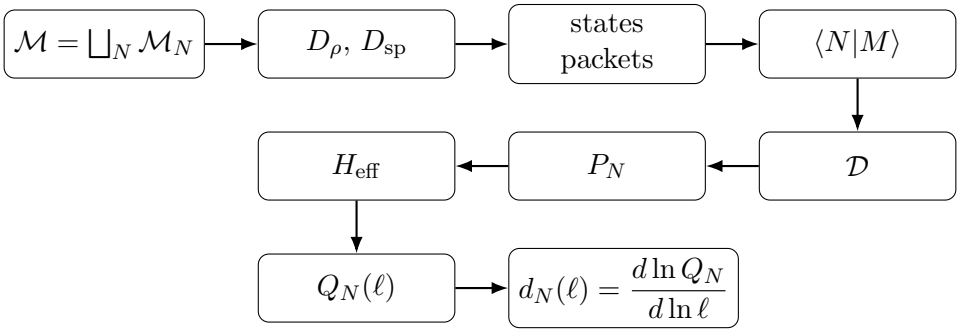


Figure 1: Frozen HSMT architecture. Dimensional characterization is downstream of the operator/effective-sector construction.

B Selector-Audit Summary

Table 2: Representative selector mechanisms and present status.

Mechanism	Present status
Λ^1–Λ^2 balance	Selects $q = 3$ if imposed; balance principle itself not derived.
Invertible $J_0 : \Lambda^2 \to V^*$	Selects $q = 3$ if imposed; exchange law remains additional premise.
Hodge/Clifford structure	Mathematically available; does not uniquely fix the physical carrier rank.
Ground-state / $\mathcal{D}^2$ selection	Does not uniquely fix rank, measure, and generator from the frozen core.
RG/symmetry stability	Can constrain candidate structures but has not supplied a unique physical selector.
Boundary/history weighting	Requires independent measure/dynamical input.
Bootstrap/self-consistency	Does not remove the need for a new physical premise in the audited constructions.

C Predeclared P3 Predictions

For the formal experiment, the theoretical predictions are fixed before the data are examined:

$$\boxed{R_S = 3} \tag{46}$$

and

$$\boxed{R_I = \sqrt{3} \approx 1.732050808.} \tag{47}$$

The preparation-screening windows are engineering criteria and must not be confused with theoretical predictions. Likewise, the final frequency-ratio comparison is to be reported with experimentally propagated uncertainty rather than with a post hoc tolerance chosen after observing the result.

References

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